Mobile Health Technologies: A Critical Review

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BACKGROUND & SIGNIFICANCE

- According to the Pew Research Center (2014), 90% of American adults have a cell phone today, and 58% of these own a smartphone
- There are more than 13,600 documented health related applications for smartphones, and this number is increasing fast (Donker, Petrie, Proudfood, Clarke, Birch, & Christensen, 2013)
- These apps target a range of foci, including diet, exercise, mental health, smoking cessation, and sobriety maintenance
- Mobile health (mHealth) technologies offer great potential, but research regarding their use and effectiveness is only beginning
- The purpose of the review presented here was to assess current mHealth literature so as to inform future implementation(s) and evaluation(s) of mHealth technologies

GUIDING QUESTIONS

1. **What is the current state of knowledge regarding mobile health applications and related technology?**
2. **What are the guiding theories and frameworks, research designs and methods, and outcome findings of published research regarding mHealth?**
3. **What are the indicated next steps for future research targeting the integration of mHealth in peoples' everyday lives?**

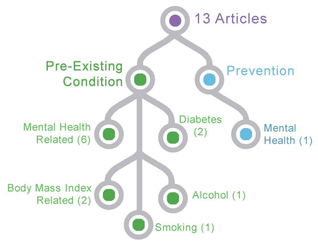
SEARCH SELECTION PROCESS

- Databases searched: Google Scholar and PubMED
- Search terms included: mHealth apps, mHealth phone applications, self-care apps, smartphone health, and combinations of the word "mobile" with apps, health apps, health, health technology, mental health and phone apps
- Articles selected by:
  - Published between 2010-2014
  - Measured effectiveness and/or use and adherence of the app
- See attached handout for list of final articles selected for review

FINDINGS

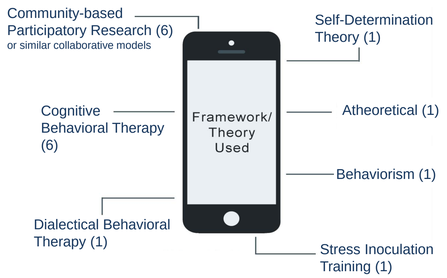
Thirteen studies representing the most recent research regarding mHealth self-care apps and mobile messaging systems were evaluated in this review. Below is a description of their principal themes and characteristics.

Intervention vs. Prevention Studies



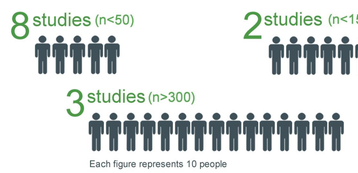
Most research focused on apps that dealt with intervention (i.e., not prevention)

Guiding Theories and Frameworks



Note: Some studies employed multiple frameworks/theories

Sample Sizes

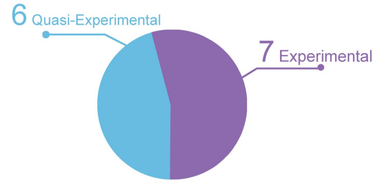


The majority of studies employed samples of less than 50 people

Studied Where?



Methods

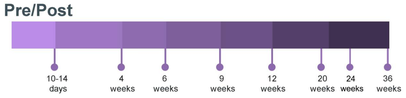


Experimental studies employed randomized assignment between control and intervention groups

Quasi-experimental studies employed a single-group repeated-measures design or lacked randomization in group assignment

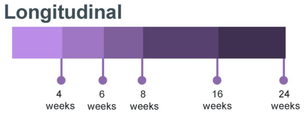
Designs

Pre/Post



Eight studies conducted pre/post design with varied timespans noted above

Longitudinal



Five studies conducted longitudinal designs with three or more data collection points

Key Findings and Outcomes

Satisfaction

- Seven studies explicitly assessed participants' satisfaction with mHealth technologies
- Results generally showed high satisfaction via self-report or focus groups
- Two studies reported that participants would recommend the app to a friend
- When dissatisfaction was reported, principal reasons included perceptions of not needing the app, technical difficulties, insufficient instruction, or low motivation

Effectiveness

- Nine studies specifically assessed effectiveness
- Results showed significant improvements in participants' health, alongside strong perceptions of app helpfulness

Adherence

- Seven studies measured adherence
- Findings showed that more than 50% of participants used the apps in the manners that investigators intended
- Two studies reported declines in adherence across four and six month follow-up
- Six studies showed adherence to app use without incentivizing participants
- Two comparison studies with websites and diaries found that high tech convenience aided in adherence to self-monitoring

DISCUSSION

- mHealth technologies target a broad range of health foci, and are becoming increasingly accessible to the lay public
- Mobile health technologies offer unique advantages over conventional health-reminder and behavior-tracking systems, including low cost, convenience of use, low- or no- stigma, and the ability for users to personalize in accord to unique needs
- Research is only beginning to understand the usability and effectiveness of mHealth, and a great deal of work remains to be done

LIMITATIONS

- Few published studies regarding mHealth have advanced longitudinal designs; understanding regarding sustained use and/or long-term outcomes is thereby limited
- Most studies regarding mHealth to-date have been under-powered, employing small samples over relatively short time periods
- Technological problems, such as loss of service or a frozen phone, can disrupt apps' usability and effectiveness
- Assessing the unique contribution of study outcomes to mHealth technology is difficult because study participants tend to use other self care methods (e.g., social support) in synchrony

FUTURE DIRECTIONS

- Future studies should employ larger samples with higher power
- Future studies should advance longitudinal designs so as to facilitate knowledge regarding sustainability of app use and effects
- Due to the specificity of each app and its audience, using Community-based Participatory Research methods could facilitate a collaborative and iterative process to create tools that are personalized, well-accepted, useful, and effective for the populations they target
- Future research should consider expanding its scope to users in high risk professions (e.g., ER physicians, graduate/medical students, trauma relief workers) as a tool to advance self care and mitigate compassion fatigue

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